

Module 3: Reflective Journal
ITAI 1371 - Machine Learning .
Neural Crew .
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This Lab allowed me to consolidate my learning. I was able to expand my understanding of ML types and Logistic Regression. I think I understand why logistic regression was a better approach for this lab, this was likely due to linear separability.

I listed concrete steps that I would take to improve the performance (like feature scaling and hyperparameter tuning). I even thought about real world applications. My field of interest is healthcare. I could see applying these concepts to a problem like predicting the likelihood of a patient developing Type 2 Diabetes.

This would be a Supervised Learning (Classification) problem. I'd need data on patient demographics, medical history, lifestyle factors, lab results (like blood glucose and HbA1c), and whether they were previously diagnosed with diabetes. The goal would be to train a model that could classify new patients as either high or low risk for developing the condition, which could help in early intervention.

To answer some of the reflection questions presented in the notebook:

Which type of machine learning interests you most and why?

Reinforcement Learning definitely interests me the most! The idea of an agent learning through trial and error by interacting with an environment, like teaching a robot to play chess or developing autonomous vehicles, is really fascinating. It feels very close to how humans learn in many situations, and the potential applications are incredibly exciting.

What was the most challenging part of the ML workflow for you?

The most challenging part for me was probably the '**Model Interpretation**' step, specifically understanding the confusion matrix and distinguishing between metrics like precision, recall, and F1-score.

It's not enough to just see an accuracy number; truly understanding why a model made certain errors and what different metrics tell you about those errors felt pretty complex initially. It requires a deeper dive than just looking at the overall accuracy.

How might you apply these concepts to a problem in your field of interest?

What questions do you have about machine learning that you'd like to explore further?

I have a few questions I'd love to explore:

1. How do you effectively handle datasets where one class is much, much larger than others (imbalanced datasets)? Like, what if only 1% of transactions are fraudulent, but you really need to catch them all?
2. What are the best strategies for selecting the most important features when you have hundreds or thousands of potential input variables?
3. How do you make sure a model doesn't just memorize the training data but can actually generalize well to completely new data, especially when you're fine-tuning a lot of parameters?